

COP2073C Module 10 Programming Assignment

For this assignment you will need to install and load the tidyverse.

In this assignment we will practice chaining functions together with pipes to create a streamlined data manipulation workflow using dplyr and select as part of mastering data manipulation in the tidyverse.

Instructions

Using the nycflights13 flights dataset, follow these steps:

1. Use pipes (`%>%`) to connect each of the following steps to create a single, streamlined pipeline of commands.
2. Use the `arrange()` function to sort the data by *earliest to latest scheduled departure time* (column `sched_dep_time`) and, for flights with the same scheduled departure time, sort by *highest to lowest delay* (column `dep_delay`) within a single call to `arrange()`.
3. Use `select()` to retrieve the columns **`sched_dep_time`, `dep_delay`, `carrier`, and `day of the month`** (the **`day`** column).

Non-Functional Requirements:

- Include a 4-line ID header at the beginning of your script.
- Include vertical spacing (a blank line) between logical blocks for readability.
- Comment your code thoughtfully (avoid excessive commenting).
- Ensure each line of code does not exceed 80 columns

Expected Output:

```
# A tibble: 336,776 × 4
  sched_dep_time dep_delay carrier    day
      <int>      <dbl> <chr>   <int>
1         106         NA US         27
2         500        188 US         24
3         500         61 US         13
4         500         47 US          9
5         500         44 US          8
6         500         29 US         17
7         500         25 US         30
8         500         22 US         28
9         500         20 US         21
10        500         18 US          3
# i 336,766 more rows
```